

? Jargon Buster

DDR RAM: Double Data Rate RAM is a type of memory that handles data at twice the speed of the old SDRAM. DDR RAM typically operates at 266 MHz, 333 MHz and 400 MHz effective speeds, while the actual speeds are 133, 166 and 200 MHz respectively.

DDR2 RAM: Double Data Rate 2 RAM is a memory type based on chips designed to run at speeds higher than that of DDR RAM, such as 533 MHz and 667 MHz. Very few motherboards right now support DDR2 RAM.

Dual-channel: When two RAM modules are plugged into two slots marked for two different memory channels, then theoretically, the data flows at twice the normal rate through the RAM. This is known as dual-channel memory configuration.

IDE: IDE stands for Integrated Drive Electronics, and is a standard interface for hard drives, optical drives and other physical storage devices. It is also known as ATA, which stands for Advanced Technology Attachment, or Parallel ATA. The various types are ATA33, ATA66, ATA100 and ATA133, based on the

transfer speed. It uses a 40-pin connector cable that may consist of 40 or 80 conductors.

PCI: Peripheral Component Interconnect is a standard used on all current-generation motherboards and expansion cards used to connect two devices. The data transfer rate of a PCI bus can be up to 133 MBps.

PCIe: Peripheral Component Interconnect Express is a new PCI standard that transfers data at speeds equivalent to AGP 16x, which is roughly 4.3 GBps.

SATA: Also abbreviated as S-ATA, short for Serial ATA, is an interface for physical storage devices such as hard drives. It is a serial link and consists of a single cable with a minimum of four wires creating a point-to-point connection between devices. The transfer rates for SATA begin at 150 MBps; for SATA II, this figure is 300 MBps.

Time-shifting: Time-shifting is a feature of PVR software that allows you to pause, replay or skip a portion of a live TV programme. With the time-shift feature, you have total control over how you watch television.

Features

Almost all the PCs here featured the Intel Pentium IV processor, but only the Datamini Festiva MG Premium sported the even more powerful Pentium IV with Hyper-Threading Technology—this puts it in a league of its own. The Acer Aspire ASP V4324 sported the AMD Athlon64 3200+, and the Krypton had a 3500+, both of which are very powerful processors.

The Datamini Festiva MG came with a CD-RW drive, and the Krypton was the lone PC with a DVD-Writer. All the others came with Combo drives. This is unfortunate, because the prices of DVD-Writers have plummeted to a level where you can get one in lieu of a Combo drive for just a few hundred rupees extra.

SATA hard drives were common in this category; the exception was the Acer Aspire SA80, which featured a slower IDE hard drive. Hard drive capacities in this category didn't see much variation; they were all either 74 or 80 GB. All the PCs had 256 MB of RAM—except for the Krypton with 512 MB—which can be considered just about OK for this price range. The funny thing is, the HCL PCs in the *value* category came with a healthy 512 MB of RAM, and we were left wondering why none of the manufacturers except Krypton could provide the same in this, the more expensive category.

The Acer ASP V4324 was the

lone PC with a GeForce4 MX 440 64 MB AGP graphics card. The others were content with onboard graphics solutions, which included the nVidia 6100, Intel GMA 900 and the SiS 330 Mirage.

PCs with the Intel GMA 900 and nVidia 6100 have the advantage of being Vista-ready, because of DirectX 9 compliance. The nVidia 6100 is also the only onboard graphics chipset that supports Shader Model 3.0. With the exception of the Compaq Presario SR1630IL, all the PCs at least provided an option to upgrade the graphics subsystem with the addition of an AGP or a PCIe graphics card.

The displays in this category were either LCDs or 17-inch CRT monitors. The Acer SA80 came with a 15-inch LCD, and the Datamini Festiva MG Premium came with a 15.4-inch widescreen LCD.

Most of the cabinets we saw were Micro ATX, with the excep-



Compaq Presario SR1630IL

tion of the two Datamini PCs, which came with ATX cabinets—thus providing four 5.25 inch bays for upgrades. The build quality of these cabinets, unfortunately, was sub-par. Air ventilation seems to have been taken into account in the design of all the cabinets, with the exception of the Krypton and the LG XD8—whose cabinets were too sleek to allow for good ventilation. The Krypton cabinet featured an eye-catching multi-coloured LED.

With the exception of the Compaq PC, all the cabinets had the USB and audio connectors on the front, enabling easier accessibility. The Acer Aspire SA80 went one step further, integrating a multi-card reader capable of reading SM/XD, SD/MMC, CF and MS/MS Pro cards. This feature is useful for those who own digital cameras or any other device that uses removable Flash cards, such as mobile phones and PDAs.

The Acer Aspire SA80 came with a wonderful package consisting of 11 CDs and DVDs: there were movies, encyclopaedias, utility software, and more. Sadly, there were no speakers bundled; Acer seems to have thought the speakers on the LCD display would do.

Most of the PCs here came with FreeDOS or with no OS at all. The exception was the Acer Aspire SA80 and the LG XD8, which had Linux installed. The Datamini Festiva MG Premium came with 5.1 speakers. The Acer Aspire ASP V4324 came with Creative 2.1 speakers and a force-feedback gamepad.

Performance

The Krypton Voyager 9, with its AMD Athlon64 3500+, was the top scorer, with the highest points in the CPU test module of SiSoft Sandra. In the real world, this means that the V4324's 64-bit processor can rip through complex computations faster than can other processors in this category.

In terms of memory module configuration, the manufacturers have done customers a disservice: for better memory bandwidth, a dual-channel configuration is recommended, with a pair of matching modules in the memory slots. All the manufacturers except Krypton left one of the two memory slots free, and if you want to upgrade to dual-channel, you'll have to procure a matching memory module from the manufacturer.